

AI-Driven Predictive Maintenance for Aerospace & Rail Fleets

Problem Statement, Use Case & Executive Summary

Problem Statement

Aerospace and rail operators sit between two costly extremes. **Scheduled maintenance** replaces or inspects components on a fixed calendar, regardless of actual wear — burning budget on parts that still have life left in them. **Reactive maintenance** waits for failure signals or inspection flags, risking unplanned downtime, safety incidents, and emergency repair costs that can run 3-5x higher than planned work.

The root cause: **structural health assessment today is siloed.**

- **Simulation engineers** run ANSYS, Abaqus, and Creo models to predict stress, fatigue, and crack growth — but these physics-based models are slow, need expert tuning, and don't self-correct against real-world sensor drift.
- **Data teams** have sensor and inspection data, but lack the physics grounding to know *why* a component is degrading, only that it might be.
- **Maintenance engineers** are stuck manually reconciling both, with no unified, explainable view — and no audit trail when regulators or safety boards ask "why was this part cleared?"

The result: fleets fly or run with **hidden uncertainty**, and the people who could close that gap (physics simulation + ML) are working from two different systems that never talk to each other.

Core question our project answers:

Can we make a physics model and a machine-learning model check each other's work automatically — and only escalate to a human when they actually disagree — so that maintenance decisions are faster, cheaper, and still trustworthy enough to sign off on?

Use Case

"From Sensor to Sign-Off" — Autonomous Fleet Health Monitoring with Human-in-the-Loop Trust

Picture a fleet engineer opening the **User Dashboard** each morning. Instead of a wall of raw sensor logs, they see fleet health scored, ranked, and explained in plain language.

Behind the scenes, here's what happened overnight:

1. **Orchestrator Agent** kicks off the analysis cycle, pulling flight logs, sensor streams, inspection reports, maintenance records, and CAD models — and recalling this component's history from **Fleet Memory**.
2. The **Digital Twin / Simulation Layer** (ANSYS, Abaqus, Creo, NASTRAN) generates a physics-based stress, fatigue, and crack-growth prediction for each critical component.
3. Two independent agents evaluate the same component in parallel:
 - The **Physics Agent** applies fatigue models, crack-growth laws, and failure physics.
 - The **ML Correction Agent** applies Random Forest, Gaussian Process, and residual-learning models trained on real historical outcomes — catching what pure physics models miss (manufacturing variance, environmental effects, sensor drift).
4. The **Judge Agent** compares both predictions, computes a confidence score, and detects divergence. If both agree — the part is cleared to continue monitoring, automatically.
5. If they *disagree*, the **Calibration Agent** re-tunes simulation parameters using Bayesian calibration, and the case is routed to a **human engineer** with a full explainable-AI report (SHAP/LIME) — not a black-box score, but the actual reasoning trail.
6. Once approved, the **Action Agent** decides the next step — continue monitoring, schedule an inspection, recommend repair, or generate a work order — and pushes it straight into **SAP PM, IBM Maximo, or AMOS**, so nothing gets lost in a spreadsheet.

Why this is compelling for judges:

- It's not "AI replaces engineers" — it's **AI agrees with itself before it bothers a human**, which is the trust model regulators and MRO teams actually want.
- It closes the loop end-to-end: raw sensor data → physics + ML cross-validation → explainable decision → ERP work order, with zero manual handoff.
- It's domain-agnostic by design — the same architecture works for aircraft structures, rail bogies, or any asset with a physics model and a sensor feed.

Executive Summary (Revised)

AI-Driven Predictive Maintenance: Physics-AI Cross-Validation for Aerospace & Rail Fleet Health

This project integrates industry-standard simulation software — **ANSYS, Abaqus, and Creo/NASTRAN** — with a multi-agent AI system that cross-validates physics-based structural predictions against machine-learning models trained on real operational data. Where the two disagree, the system automatically recalibrates and escalates to a human engineer with a full explainable-AI report; where they agree, maintenance decisions proceed autonomously.

By combining fatigue-life models — **Basquin's equation, Paris' Law, and NASGRO** — with ML residual correction (Random Forest, Gaussian Process regression), the system achieves a measurable accuracy improvement in stress and fatigue-life prediction over physics-only models, while keeping every decision traceable back to both the simulation and the data that informed it.

The result is a fleet health system that is faster than manual review, more accurate than physics-only simulation, and transparent enough for engineers and regulators to trust — feeding directly into SAP PM / IBM Maximo for real-world action.

Suggested next tightening for your report

- Replace "3%+ accuracy improvement" with the actual delta you can defend live (judges will ask for the baseline comparison — physics-only RMSE vs. physics+ML RMSE).
- Add one concrete before/after number to the use case (e.g., "reduced false-positive inspection flags by X%" or "cut mean time-to-decision from N hours to M minutes") — a single hard number does more for judges than the full architecture explanation.